

Timing, Outcome, and Risk Factors of Intracranial Hemorrhage in Acute Respiratory Distress Syndrome Patients During Venovenous Extracorporeal Membrane Oxygenation

OBJECTIVES: Intracranial hemorrhage is a serious complication in patients receiving venovenous extracorporeal membrane oxygenation during treatment of the acute respiratory distress syndrome. We analyzed timing, outcome, and risk factors of intracranial hemorrhage in patients on venovenous extracorporeal membrane oxygenation.

DESIGN: Retrospective cohort study.

SETTING: Single acute respiratory distress syndrome referral center.

PATIENTS: Patients receiving venovenous extracorporeal membrane oxygenation were identified from a cohort of 1,044 patients with acute respiratory distress syndrome. Patients developing an intracranial hemorrhage during venovenous extracorporeal membrane oxygenation therapy were compared with patients without evidence for intracranial hemorrhage. The primary objective was to assess the association of intracranial hemorrhage with 60-day mortality. Further objectives included the identification of risk factors for intracranial hemorrhage and the evaluation of clinical cutoff values.

INTERVENTIONS: None.

MEASUREMENTS AND MAIN RESULTS: Among 444 patients treated with venovenous extracorporeal membrane oxygenation, 49 patients (11.0% [95% CI, 8.3–14.4%]) developed an intracranial hemorrhage. The median time to intracranial hemorrhage occurrence was 4 days (95% CI, 2–7 d). Patients who developed an intracranial hemorrhage had a higher 60-day mortality compared with patients without intracranial hemorrhage (69.4% [54.4–81.3%] vs 44.6% [39.6–49.6%]; odds ratio 3.05 [95% CI, 1.54–6.32%]; $p = 0.001$). A low platelet count, a high positive end expiratory pressure, and a major initial decrease of Paco_2 were identified as independent risk factors for the occurrence of intracranial hemorrhage. A platelet count greater than 100/nL and a positive end expiratory pressure less than or equal to 14 cm H_2O during the first 7 days of venovenous extracorporeal membrane oxygenation therapy as well as a decrease of Paco_2 less than 24 mm Hg during venovenous extracorporeal membrane oxygenation initiation were identified as clinical cutoff values to prevent intracranial hemorrhage (sensitivity 91% [95% CI, 82–99%], 94% [85–99%], and 67% [48–81%], respectively).

CONCLUSIONS: Intracranial hemorrhage occurs early during venovenous extracorporeal membrane oxygenation and is a determinant for 60-day mortality. Appropriate adjustment of identified modifiable risk factors might lower the prevalence of intracranial hemorrhage during venovenous extracorporeal membrane oxygenation therapy.

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Venovenous extracorporeal membrane oxygenation (VV-ECMO) is increasingly used in patients with the severe acute respiratory distress syndrome (ARDS) (1, 2). VV-ECMO secures gas exchange and enables lung-protective ventilation (3). Morbidity and mortality in patients with VV-ECMO remains high and is significantly influenced by complications arising from the treatment with VV-ECMO itself (4, 5). Although technical complications such as oxygenator failing are associated with the ECMO system itself, common medical complications include blood stream infections, thromboembolisms, and bleeding (4).

Intracranial hemorrhage (ICH) occurs in 4–16% of ARDS patients treated with VV-ECMO (6–9). Although ICH is a serious complication, data on outcome of these patients are still limited and solely originate from small retrospective studies. Although there is evidence of a higher mortality in patients developing ICH during VV-ECMO therapy, previous studies have not evaluated ICH as an independent determinant for mortality (6, 9).

Although prevention of ICH is crucial, data on timing and predisposing factors of ICH in ARDS patients treated with VV-ECMO are rare. Previous findings suggested that ICH occurs most frequently during the first week after VV-ECMO initiation (9). An independent association with the occurrence of ICH during VV-ECMO therapy has been demonstrated for renal failure, thrombocytopenia, and a rapid decrease of the Paco_2 during ECMO initiation (6, 9). So far, none of these risk factors has been confirmed in other cohorts of ARDS patients treated with VV-ECMO.

Using a large cohort of ARDS patients treated with VV-ECMO, the objective of this study was to provide further evidence on timing and outcome of ICH during therapy with VV-ECMO and to identify risk factors including clinical cutoff values that might contribute to clinical decision-making for ICH prevention.

METHODS

Study Design and Setting

This is a retrospective cohort study of ARDS patients treated with VV-ECMO at the ARDS referral center of the Department of Anesthesiology and Intensive

Care Medicine, Charité – Universitätsmedizin Berlin, Campus Virchow-Klinikum, Berlin, between January 2007 and December 2018. Patients with ICH during treatment with VV-ECMO were compared with patients without evidence for an ICH. After assessing occurrence rates and timing of ICH, the association of ICH with 60-day mortality was evaluated. Risk factors for ICH during VV-ECMO were identified. The study was approved by the Medical Ethics Committee of Charité – Universitätsmedizin Berlin (No. EA1/007/19).

Participants

All adult patients who fulfilled the criteria of the Berlin Definition for ARDS (10) and who were treated with ECMO were eligible for the study. Patients were excluded if they: 1) had ICH before ECMO initiation, 2) received venoarterial ECMO, 3) were converted from venovenous to venovenous arterial ECMO during ARDS treatment, and 4) had incomplete datasets to evaluate the primary endpoint.

Assessment of ICH and Grouping

All patients were evaluated with a baseline cranial CT (CCT) scan before or immediately after the start of VV-ECMO therapy. Patients who received VV-ECMO implantation by our ECMO retrieval team in a referring hospital received a baseline CCT scan at admission to our center (generally within 2 hr after ECMO implantation). During VV-ECMO therapy, neurologic examinations were performed 8 hourly by the treating intensive care physician and 4 hourly by the nurses in charge. A follow-up CCT scan was performed if a new neurologic symptom was observed such as mydriasis, anisocoria, seizures, or persistent coma on sedation withdrawal.

All ICHs were diagnosed by an expert neuroradiologist. The acronym ICH is most commonly used for the term “intracerebral hemorrhage or intraparenchymal hemorrhage”. To ensure comparability with previous studies (6–8, 11), ICH was referred as to “intracranial hemorrhage” in this study and was defined as occurrence of parenchymal hemorrhage, nonaneurysmal subarachnoid hemorrhage (SAH), or subdural hematoma (SDH). We considered all ICHs irrespectively of size and location. Patients were grouped into a patient cohort with evidence of ICH and a patient cohort without evidence of ICH. Because previous data indicated that the majority of patients develop ICH during

the first week after ECMO initiation (9), patients with an ICH were further stratified into early ICH (ICH within the first 7 d after ECMO initiation) and late ICH (ICH after 7 d after ECMO initiation).

Patient Management During ECMO

Blood and platelet counts, prothrombin time, and the activated partial thrombin time (aPTT) were routinely measured 8 hourly. Unless contraindicated, all patients received an initial bolus followed by a continuous infusion of unfractionated heparin targeting an aPTT of 50–75 seconds. Further details on coagulation management, the management of the membrane oxygenator, and ECMO circuit are provided in the **Online Data Supplement** (Supplementary Digital Content 1, <http://links.lww.com/CCM/G47>).

Objectives, Bias Handling, and Data Sources

The primary objective was to assess the association of the occurrence of ICH during VV-ECMO with the mortality within 60 days after ARDS onset. To address a potential selection bias, prognostic baseline characteristics and variables of the validated Respiratory ECMO Survival Prediction (RESP) Score were considered (12). Secondary objectives included the evaluation of timing of ICH, the identification of risk factors for the occurrence of ICH, and their evaluation with respect to clinical cutoff values. All data required for this study were extracted from the two electronic patient data management systems in use at the hospital (COPRA, Sasbachwalden, Germany; and SAP, Walldorf, Germany). Further details are provided in the Online Data Supplement (Supplementary Digital Content 1, <http://links.lww.com/CCM/G47>).

Statistical Analyses

Differences of continuous data were tested using the exact Mann-Whitney *U* test and the Kruskal-Wallis test, and frequencies were tested using Fisher exact test. To evaluate the association of the occurrence of ICH with mortality, a multivariable logistic regression was performed using a backward variable selection procedure based on the Akaike information criterion (AIC).

To evaluate potential risk factors as precise as possible, variables of gas exchange, hemostatic variables, and ventilator settings were compared between patients with and without ICH within defined periods: day of

ECMO initiation, the first 7 days after ECMO initiation, and the period after the first 7 days after ECMO initiation. Hemostatic variables were processed for each period using an average of daily minimum (platelets and fibrinogen) or maximum (prothrombin time) values. All other variables of interest were processed using average or maximum values, as appropriate. The identification of independent risk factors for ICH was performed separately for early and late ICH using two separate multivariable logistic regression models with a backward variable selection procedure based on the AIC. Baseline characteristics and variables of interest that were different in univariable analysis were included, as appropriate. Clinical cutoffs of risk factors were identified using a gray zone approach (13). The gray zone technique proposes two cutoffs: a high sensitivity cutoff that allows exclusion of the diagnosis with near certainty (no ICH), and a high specificity cutoff that includes the diagnosis with near certainty (occurrence of ICH). A sensitivity of 90% and a specificity of 90% were chosen, respectively. The best cutoff according to Youden was calculated if the gray zone approach could not be applied (14). Due to the exploratory study type, all analyses were considered to be nonconfirmatory. A two-tailed *p* value of less than 0.05 was considered statistically significant. The statistical analyses were performed with the use of R software, Version 3.6.1 (R Project for Statistical Computing, Vienna, Austria).

RESULTS

A total of 479 ARDS patients with ECMO were identified. Of those, 35 patients were excluded, so that a total of 444 patients with VV-ECMO were eligible for analysis (**Fig. 1**).

Forty-nine patients (11.0% [95% CI, 8.3–14.4%]) developed ICH on VV-ECMO treatment during ARDS therapy with 33 patients having early ICH (7.4% [5.2–10.3%]) and 16 patients having late ICH (3.6% [2.1–5.9%]). The median time to occurrence of ICH since ECMO initiation was 4 days (95% CI, 2–7%; range, 0–32) (**Fig. 2A**). Parenchymal hemorrhage occurred in 38 patients (77.6% [63–87%]), nonaneurysmatic SAH in 11 patients (22.4% [12–37%]), and SDH in one patient (who had simultaneously a SAH and was therefore classified as a patient with SAH). In 32 patients, the ICH was an incidental finding detected during follow-up CCT scans (65.3% [50–78%]), whereas clinical neurologic symptoms such as anisocoria or mydriasis

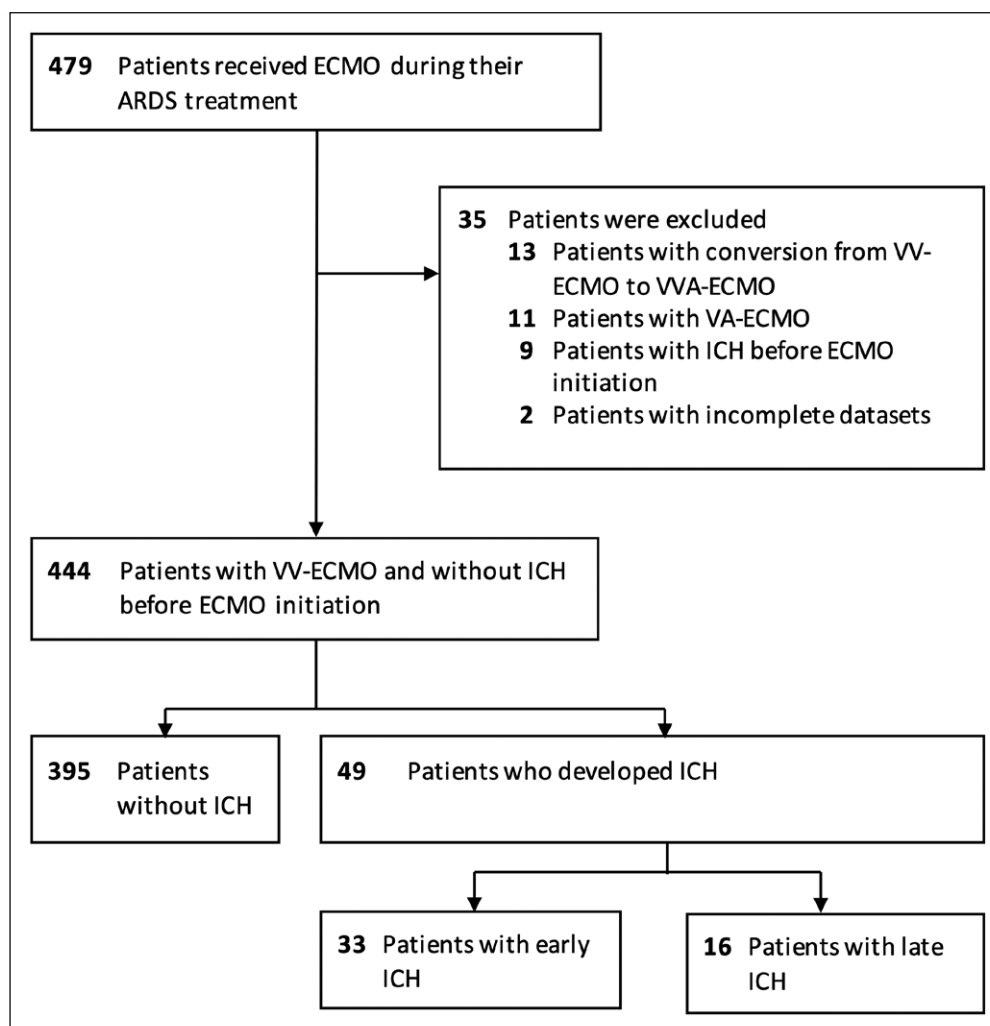


Figure 1. Study flow diagram. ARDS = acute respiratory distress syndrome, ECMO = extracorporeal membrane oxygenation. ICH = intracranial hemorrhage, VA = venoarterial, VV = venovenous, VVA = venovenous arterial.

appeared in 16 patients (32.6% [20–48%]). The majority of ICHs was located supratentorial (35 patients; 81.4% [66–91%]) (Table E1, Supplementary Digital Content 2, <http://links.lww.com/CCM/G48>). A total of 73.4% (95% CI, 68.7–77.6%) of the patients that were classified as patients without evidence for an ICH had a CCT scan during the course of VV-ECMO therapy. Median time to the CCT scan was 14 days (IQR, 3–27 d).

Baseline characteristics of patients without and with ICH are shown in Table E2 (Supplementary Digital Content 3, <http://links.lww.com/CCM/G49>). There were no differences in age, gender, comorbidities, admissions scores, ARDS severity, ARDS etiology, and time of ECMO initiation. Patients with an early ICH presented more often with septic shock, a platelet count less than 50/nL, and required renal replacement therapy at ECMO initiation and have had more

frequently previous chemotherapy. Patients with a late ICH had similar baseline characteristics compared with patients without occurrence of an ICH.

Outcome

Within 60 days, patients who developed an ICH during VV-ECMO had a higher mortality compared with patients without ICH (69.4% [95% CI, 54–81%] vs 44.6% [39–49%]; odds ratio [OR], 2.82 [95% CI, 1.51–5.48]; $p = 0.001$) (Fig. 2B). After adjusting for septic shock and RESP score variables, ICH was independently associated with a three-fold increased chance for death within 60 days after ECMO initiation (OR, 3.05 [1.54–6.32]; $p = 0.001$) (Fig. 2B) (Table E3, Supplementary Digital Content 4, <http://links.lww.com/CCM/G50>).

Although there was no difference in 60-day mortality between patients with early and late ICH (66.6% [48–81%] vs 75.0% [57–91%]; $p = 0.79$), both, early and late ICH, were independent determinants for death within 60 days after ECMO initiation (OR, 2.71 [1.20–6.39]; $p = 0.017$ and OR, 3.98 [1.23–15.5]; $p = 0.028$).

Risk Factors

Patients with ICH had a greater decrease in $Paco_2$ and greater increase of pH during ECMO initiation compared with patients without ICH, whereas there was no difference in change of the Pao_2 . As expected, there was a strong inverse correlation between changes of $Paco_2$ and changes of pH. Patients with early and late ICH had lower platelet counts and were ventilated with higher peak inspiratory pressure, plateau pressure,

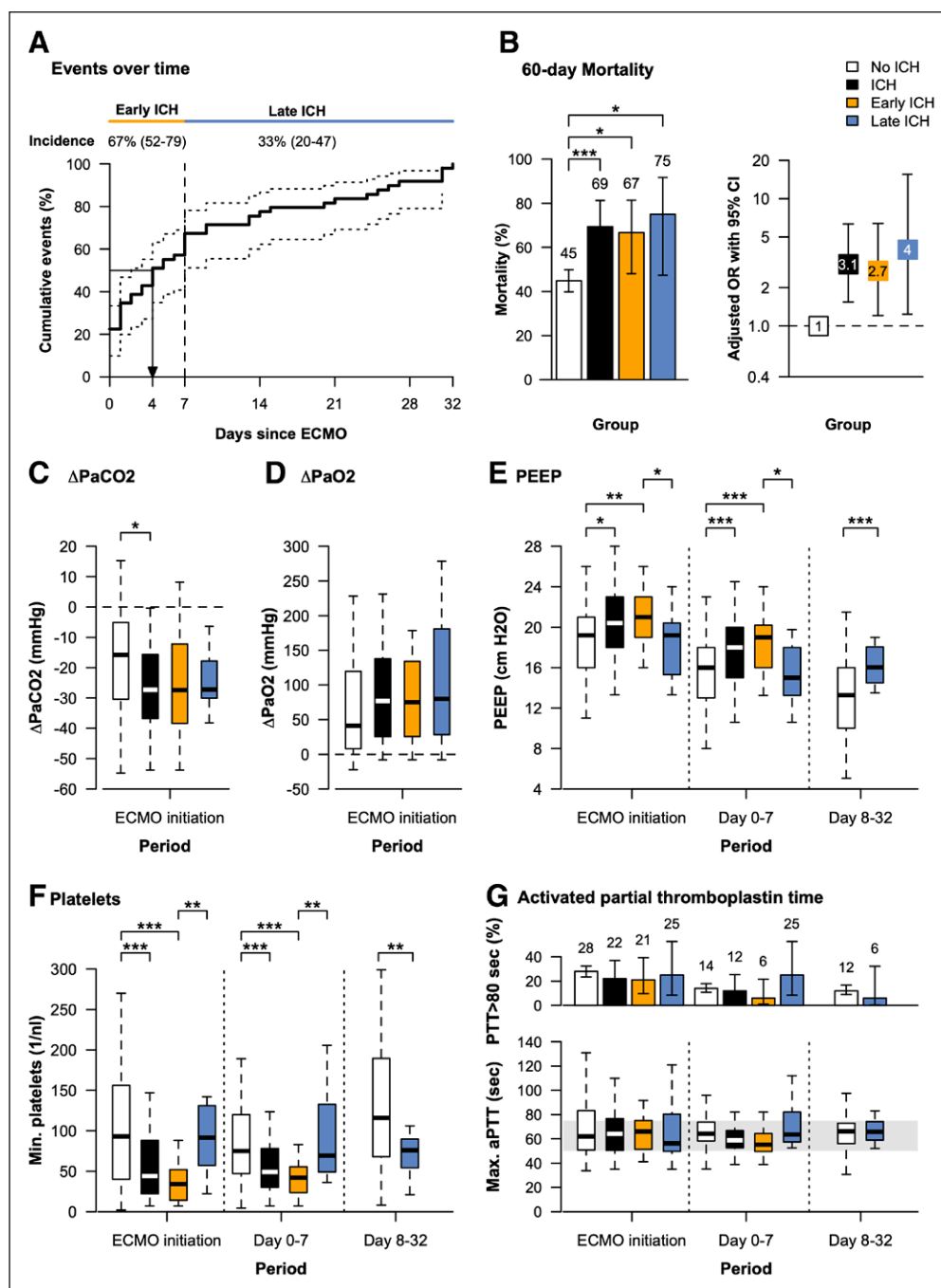


Figure 2. Timing of intracranial hemorrhage (ICH), 60-d mortality, and variables for risk factor evaluation. Cumulative events of ICH (with 95% CI) during venovenous extracorporeal membrane oxygenation (ECMO) are presented (**A**). Periods of early and late ICH, and the median time of ICH diagnosis (arrow) are indicated. Crude 60-d mortality with 95% CIs and corresponding odds ratio (ORs) with 95% CIs after adjusting for Respiratory ECMO Survival Prediction score variables are shown (**B**). Variables for risk factor evaluation are shown in **C–G**. In ICH patients, only measurements before the ICH were considered. Absolute changes of $Paco_2$ ($\Delta PaCO_2$) and Pao_2 (ΔPaO_2) were defined as post-ECMO initiation values—pre-ECMO initiation values. Data from $n = 205$ patients were available to calculate $\Delta PaCO_2$ and ΔPaO_2 . The gray box (**G**) visualizes the target for activated partial thrombin time (aPTT). max = maximum, min = minimum, PEEP = positive end expiratory pressure, PTT = partial thrombin time.

and positive end expiratory pressure (PEEP) levels during the course of VV-ECMO therapy compared with patients without ICH. However, patients with a late ICH had similar platelet counts and ventilator settings as patients without an ICH during the first 7 days of VV-ECMO therapy. There were minor differences in prothrombin time during the first 7 days of VV-ECMO therapy, but no differences in anticoagulation and fibrinogen levels during the course of VV-ECMO therapy (**Fig. 2C–G**) (**Fig. E1**, Supplementary Digital Content 5, <http://links.lww.com/CCM/G51>; **Fig. E2**, Supplementary Digital Content 6, <http://links.lww.com/CCM/G52>; and **Fig. E3**, Supplementary Digital Content 7, <http://links.lww.com/CCM/G53> [**legend**, Supplementary Digital Content 1, <http://links.lww.com/CCM/G47>]).

Three independent risk factors for early ICH were identified: a major decrease of $Paco_2$ during ECMO initiation, low platelets (including platelet count at day of ECMO initiation), and a high PEEP during the first 7 days of VV-ECMO. Low platelet counts and a high PEEP after the first 7 days of VV-ECMO were identified as independent risk factors for the occurrence

TABLE 1.

Assessment of Risk Factors for Occurrence of Early and Late Intracranial Hemorrhage During Venovenous Extracorporeal Membrane Oxygenation in Univariable and Multivariable Analysis

Model and Variables	Univariable Model, OR (95% CI)	<i>p</i>	Multivariable Model ^a , OR (95% CI)	<i>p</i>
Model 1: early ICH				
Previous chemotherapy	3.39 (1.34–7.84)	0.005	–	
Septic shock at ECMO initiation	4.48 (1.72–15.3)	0.005	–	
Renal replacement therapy at ECMO initiation	2.56 (1.17–6.20)	0.024	–	
pH at ECMO initiation (per 0.1)	0.65 (0.49–0.87)	0.003	–	
Decrease of Paco ₂ during ECMO initiation (per 10 mm Hg) ^b	1.15 (1.01–1.33)	0.031	1.22 (1.05–1.44)	0.010
PIP at ECMO initiation (per cm H ₂ O) ^b	1.06 (1.02–1.10)	< 0.001	–	
PEEP at ECMO initiation (per cm H ₂ O)	1.06 (1.02–1.12)	0.004	–	
Platelets at ECMO initiation (per 10/nL) ^c	0.82 (0.74–0.90)	< 0.001	–	
PIP day 0–7 (per cm H ₂ O)	1.16 (1.08–1.24)	< 0.001	–	
PEEP day 0–7 (per cm H ₂ O)	1.23 (1.12–1.36)	< 0.001	1.28 (1.11–1.52)	0.001
Platelets day 0–7 (per 10/nL)	0.78 (0.68–0.87)	< 0.001	0.83 (0.71–0.94)	0.012
Prothrombin time day 0–7 (per 10%)	0.79 (0.64–0.98)	0.033	–	
Model 2: late ICH				
PIP day 8–32 (per cm H ₂ O)	1.06 (1.02–1.10)	< 0.001	–	
PEEP day 8–32 (per cm H ₂ O)	1.21 (1.07–1.38)	0.001	1.15 (1.01–1.32)	0.033
Platelets day 8–32 (per 10/nL)	0.87 (0.77–0.95)	0.011	0.89 (0.79–0.98)	0.045

ECMO = extracorporeal membrane oxygenation, ICH = intracranial hemorrhage, OR = odds ratio, PEEP = positive end expiratory pressure, PIP = peak inspiratory pressure.

^aMultivariable logistic regression analyses using a backward variable selection procedure based on the Akaike information criterion (AIC). According to AIC analyses, the following variables were removed from model 1: previous chemotherapy, septic shock, renal replacement therapy, pH, PIP at ECMO initiation, PIP day 0–7, PEEP at ECMO initiation, and prothrombin time. The following variables were removed from model 2: PIP day 8–32. Internal validation of the models demonstrated a good discrimination for model 1 (early ICH) and a fair discrimination for model 2 (late ICH), respectively ($c = 0.83$ [95% CI, 0.75–0.91] and $c = 0.79$ [0.72–0.85]).

^bDue to the strong correlation of Paco₂ and pH, only Paco₂ was considered in the analysis. Due to the strong correlation of PIP and plateau pressure, only PIP was considered in the analysis.

^cPlatelets at ECMO initiation showed similar results (adjusted OR, 0.86 [0.76–0.95]; $p = 0.003$) as platelets day 0–7 and did not change the backward variable selection procedure in the analysis. Only platelets day 0–7 are presented in the article to enhance clarity of the results. A schematic of the two model is available from Figure E2 (Supplementary Digital Content 6, <http://links.lww.com/CCM/G52>).

of late ICH (**Table 1**) (Fig. E3, Supplementary Digital Content 7, <http://links.lww.com/CCM/G53> [legend, Supplementary Digital Content 1, <http://links.lww.com/CCM/G47>]). Internal validation of the models demonstrated a good discrimination for early ICH and a fair discrimination for late ICH, respectively

($c = 0.83$ [95% CI, 0.74–0.90]) and $c = 0.79$ [0.72–0.85]). Due to the strong correlation, changes of Paco₂ and changes of the pH were interchangeable in the regression analyses. Paco₂ was included for further clinical cutoff determination because Paco₂ is the determinant that is primarily affected by VV-ECMO.

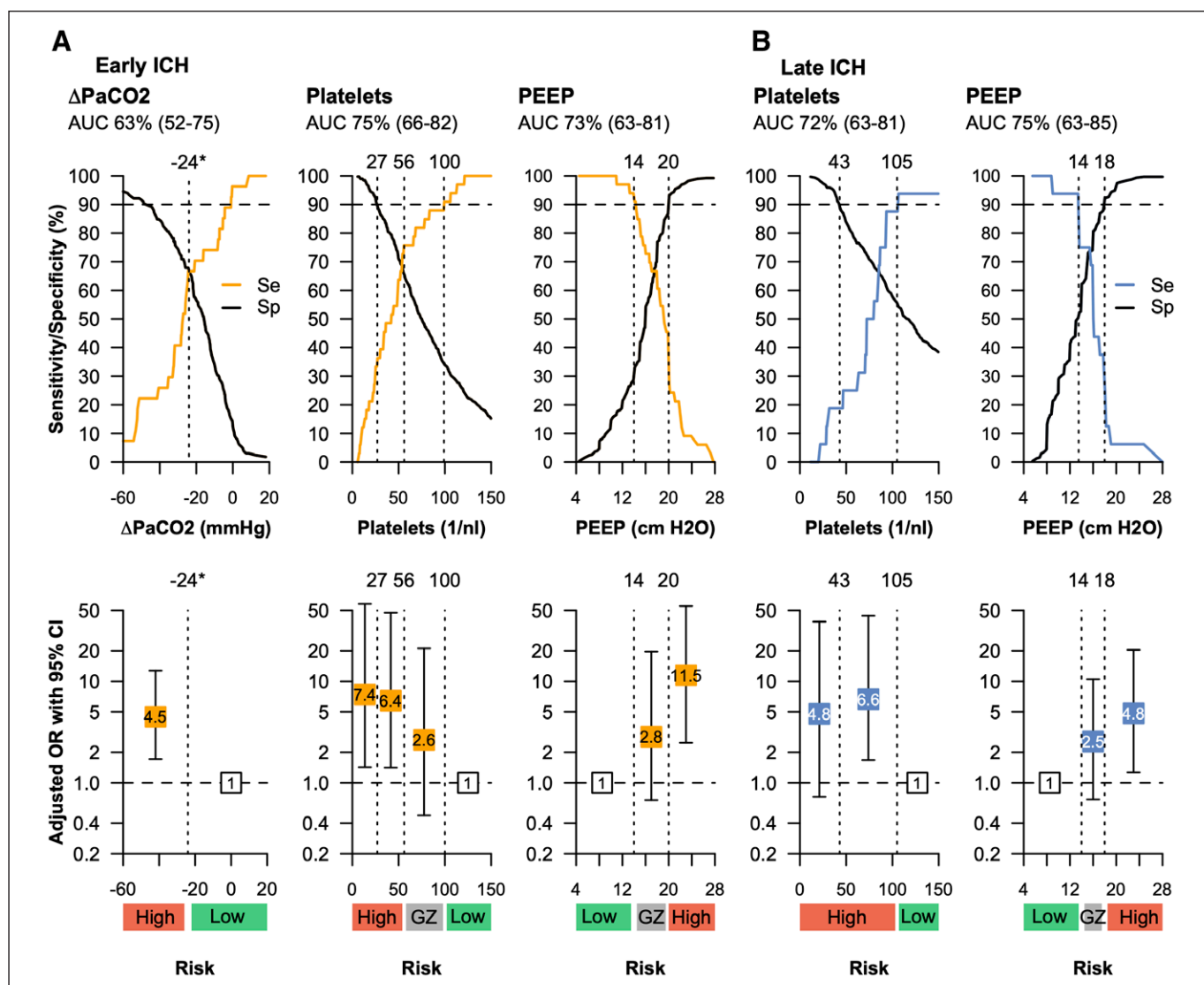


Figure 3. Sensitivity/specificity analyses for clinical cutoff determination of independent risk factors for early (A) and late (B) intracranial hemorrhage (ICH) during venovenous extracorporeal membrane oxygenation. Except for the change of PaCO_2 (*best cutoff according to Youden), cutoffs with 90% sensitivity and 90% specificity are indicated. The area under the curve (AUC) with 95% CI for each risk factor is presented. The lower panels show the corresponding chances (odds ratios) for ICH after grouping the population according the clinical cutoffs. The odds ratios (ORs) were adjusted for all independent risk factors and are presented on a log scale. GZ = gray zone, PEEP = positive end expiratory pressure.

Independent risk factors for early and late ICH with clinical cutoff values and corresponding chances for ICH are shown in **Figure 3**. A single cutoff was provided for the change of PaCO_2 during ECMO initiation because a cutoff with 90% sensitivity (decrease of $\text{PaCO}_2 < 1$ mm Hg) is not feasible for clinical practice. A decrease in PaCO_2 of greater than 24 mm Hg was identified as the best cutoff value according to Youden (sensitivity 66% [76–97%], specificity 67% [82–99%], respectively). Patients with a decrease in PaCO_2 greater than 24 mm Hg had a greater than four-fold risk for early ICH (adjusted OR, 4.47 [1.7–12.7]; $p = 0.003$)

compared with patients with a decrease in PaCO_2 less than 24 mm Hg (**Fig. 3A**).

An average minimum platelet count greater than 100/nL and an average PEEP less than or equal to 14 cm H₂O during the first 7 days of VV-ECMO were identified as cutoffs to prevent patients from developing early ICH with near certainty (sensitivity 90.9% [95% CI, 82–99%] and 93.9% [85–99%], respectively). In contrast, patients with an average minimum platelet count less than or equal to 27/nL and an average PEEP greater than 20 cm H₂O developed an early ICH with near certainty (specificity 92% [89–94%], and 93% [90–96%], respectively).

With regard to a relevant clinical perspective, an average minimum platelet count greater than 56/nL was identified as further cutoff to prevent most patients from early ICH occurrence (sensitivity 75.7% [61–91%]) and was therefore included into analyses (Fig. 3A).

An average minimum platelet count greater than 105/nL and an average PEEP less than or equal to 14 cm H₂O during the period after the first 7 days after ECMO initiation were identified as cutoffs for prevention of a late ICH with near certainty (sensitivity 93.7% [95% CI, 81–99%], and 93.7% [82–99%], respectively), whereas an average minimum platelet count less than or equal to 42/nL and an average PEEP greater than 18 cm H₂O were identified as cutoffs for occurrence of a late ICH with near certainty (specificity 90.1% [87–93%], and 92.3% [89–95%], respectively) (Fig. 3B).

DISCUSSION

Analyzing a large cohort of ARDS patients treated with VV-ECMO, we found an prevalence of 11.0% [95% CI, 8.3–14.4%]) for ICH occurrence during VV-ECMO therapy. The majority of patients developed ICH during the first week after ECMO initiation, and parenchymal hemorrhage was the predominant bleeding type. We found that ICH was an independent and strong determinant for 60-day mortality with a three-fold increased risk of death. A low platelet count, a high PEEP, and a major decrease of PaCO₂ at ECMO initiation were identified as independent risk factors for the occurrence of ICH.

The prevalence of ICH reported here is in line with previous data in patients treated with VV-ECMO ranging from 4% to 16% (6–9). Our results confirm previous findings on the timing of an ICH demonstrating that the first week after ECMO initiation represents the most vulnerable phase for ICH occurrence (9), whereas thereafter an ICH occurs only sporadically. Parenchymal hemorrhage represented almost 80% of the bleedings, which is in agreement with a recent study reporting a parenchymal hemorrhage rate of 71% (6). In most patients, ICH was located supratentorial, whereas an isolated infratentorial ICH was very rare.

Our data demonstrate that ICH during VV-ECMO therapy is an independent determinant for mortality. Even though ICH is an early complication during VV-ECMO therapy, our results show that mortality is irrespective of the timing of ICH. The 69% mortality observed in our patients with ICH is in accordance

with mortality rates of 70% found by Luyt et al (9). However, there were two other studies in ICH patients, reporting mortality rates of 32% and 34%, respectively (6, 7). These inconsistencies might be due to a lower ARDS severity with very low mortality rates of 27% and 26% among the non-ICH patients, respectively. Furthermore, both studies reported only a small mortality difference between non-ICH and ICH patients, suggesting a low ICH severity.

In concordance with Luyt et al (9), this study confirmed that a major decrease of PaCO₂ at ECMO initiation was independently associated with the occurrence of an ICH (9). Patients with a PaCO₂ decrease greater than 24 mm Hg had a more than four-fold chance for ICH. The cutoff identified in this study was concordant to the cutoff reported by Luyt et al (9) (decrease of PaCO₂ > 27 mm Hg). Changes in PaCO₂ have a major impact on cerebral physiology, including cerebral blood flow and metabolic activity (15). A decrease of PaCO₂ causes cerebral vasoconstriction, which reduces cerebral blood flow and decreases the cerebral oxygen supply. Also, a decrease of PaCO₂ can enhance neuronal excitability and thus increase the cerebral metabolic rate of oxygen (16). Taken together, the two effects might result in transient, local cerebral hypoxia. The initial rapid decrease of PaCO₂ may be followed by a rebound hyperemia and reperfusion injury to previously hypoxic brain regions (16–18) representing a putative cause for the development of an ICH. Although the exact pathomechanism is still not completely understood, data suggest that a major decrease of PaCO₂ at ECMO initiation should be avoided.

A low platelet count was associated with ICH occurrence during VV-ECMO therapy. During the first week of VV-ECMO therapy, two clinically important cutoffs were identified. Patients with a daily minimum platelet count above the high sensitivity cutoff at greater than 100/nL most likely did not develop an ICH during therapy with VV-ECMO. Our data suggest that even a daily minimum platelet count up to a lower limit of 56/nL can be tolerated, whereas values less than 56/nL were associated with a significantly increased risk for ICH. So far, only one other study detected an association between thrombocytopenia and the occurrence of an ICH (6). In the current study, we evaluated the average daily minimum value of the platelet count. This approach might retrace the clinical course including implications and measures taken by the treating

physicians. Previous studies might have failed to detect this effect by considering solely the platelet count before ECMO initiation or by comparing only the lowest platelet counts during the entire ECMO therapy (6, 9).

With regard to ventilator settings, a high PEEP was identified as a risk factor for the occurrence of an ICH during VV-ECMO therapy. In patients with brain injury but without lung injury, the application of PEEP has no effect on intracranial pressure (ICP) or cerebral perfusion pressure (CPP) (19, 20). However, a recent study in brain-injured patients showed that in patients with accompanying severe lung injury, the application of PEEP had a significant but clinically nonrelevant effect on ICP and CPP (21). The interaction between PEEP and cerebral hemodynamics may be influenced by multiple patient-specific factors (22). In this regard, a severely reduced lung compliance or the frequent use of prone positioning might contribute to ICH occurrence during VV-ECMO therapy. However, this correlation needs confirmation by further studies.

This study has several limitations. Follow-up CCT scans were not performed routinely in all patients during the course of ARDS treatment. Therefore, we cannot rule out that patients without ICH suffered from a nondetected, clinically asymptomatic ICH during VV-ECMO therapy. However, 73.4% of the patients who were classified as patient without evidence for an ICH had a CCT scan during the course of VV-ECMO therapy. Thus, the potential bias that patients might have suffered from a nondetected ICH is presumably quite low. Although patients who presented with a new neurologic symptom and in whom a SAH was detected underwent CT angiography to exclude aneurysms and other vascular bleeding sources, patients in whom a SAH was detected as incidental finding during routine CT scans did not undergo CT angiography. Due to atypical bleeding patterns in these patients, a vascular origin was considered unlikely, and therefore, contrast-enhanced imaging for assessment of cerebral vasculature was omitted.

Although the standards for ventilation and anticoagulation management during VV-ECMO therapy at our center comply with current recommendations, generalizing the clinical cutoffs of the identified risk factors is problematic because other ECMO centers might differ in their patient, device, and anticoagulation management (23). In absence of an additional advanced coagulation testing, we cannot exclude that acquired disorders of primary or secondary hemostasis

might have contributed to the occurrence of ICH. Furthermore, regression analyses cannot prove a direct cause-effect relationship. Although a causative effect of a low platelet count for ICH is widely accepted, it remains unknown if the relationship of change of Paco_2 and PEEP levels with occurrence of ICH is potentially causative or merely coincidental.

In conclusion, ICH occurs early during VV-ECMO and is a strong determinant for 60-day mortality. A low platelet count, a high PEEP, and a major initial decrease of Paco_2 were identified as independent risk factors. Appropriate adjustment of these three risk factors might prevent ICH during VV-ECMO therapy.

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The study was approved by the Medical Ethics Committee of Charité-Universitätsmedizin Berlin (No. EA1/007/19).

Data are available from the corresponding author.

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Dr. Menk contributed to conception and design and study supervision. Drs. Hunsicker, Beck, Graw, and Menk contributed to acquisition of data. Drs. Hunsicker, Beck, Weber-Carstens, Graw, and Menk contributed to interpretation of data. Drs. Hunsicker, Krannich, and Menk contributed to statistical analysis. Drs. Hunsicker, Graw, and Menk contributed to drafting of the article. All authors contributed to critical revision of the article for important intellectual content. Drs. Hunsicker, Weber-Carstens, Spies, Graw, and Menk contributed to final revision of article.

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